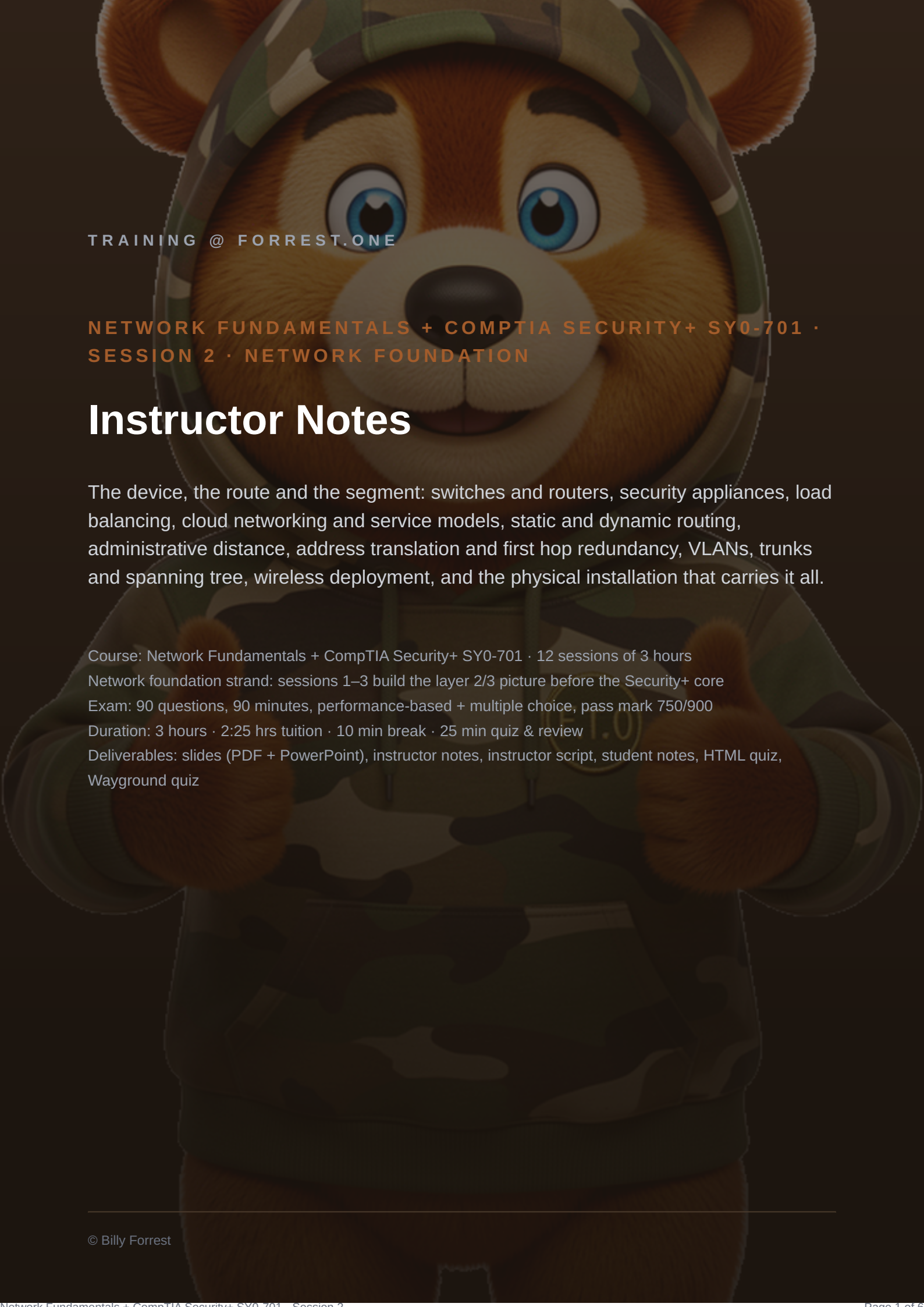




TRAINING @ FORREST.ONE

**NETWORK FUNDAMENTALS + COMPTIA SECURITY+ SY0-701 ·
SESSION 2 · NETWORK FOUNDATION**

Instructor Notes

The device, the route and the segment: switches and routers, security appliances, load balancing, cloud networking and service models, static and dynamic routing, administrative distance, address translation and first hop redundancy, VLANs, trunks and spanning tree, wireless deployment, and the physical installation that carries it all.

Course: Network Fundamentals + CompTIA Security+ SY0-701 · 12 sessions of 3 hours

Network foundation strand: sessions 1–3 build the layer 2/3 picture before the Security+ core

Exam: 90 questions, 90 minutes, performance-based + multiple choice, pass mark 750/900

Duration: 3 hours · 2:25 hrs tuition · 10 min break · 25 min quiz & review

Deliverables: slides (PDF + PowerPoint), instructor notes, instructor script, student notes, HTML quiz, Wayground quiz

Instructor Notes — Switching, VLANs & Network Architectures

A 3-hour delivery guide for Session 2. This session is visual and practical: draw the switch, the forwarding table, the routing table, the VLAN plan and the spanning tree election on the whiteboard. The class should leave able to place a device on the right layer, trace a frame through a switch, explain how a route is chosen, and justify segmentation and zoning as controls rather than plumbing.

Session at a glance

Time	Block	Slides	Focus / activity
0:00–0:10	Welcome & objectives	1–2	Where we are; the twenty outcomes
0:10–0:50	Infrastructure devices	3–9	Switching against routing, forwarding table, appliances, proxies, storage; discussion
0:50–1:15	Cloud and routing foundations	10–16	Virtualisation, cloud models, static and dynamic routing, OSPF
1:15–1:45	Route selection and translation	17–23	EIGRP and BGP, administrative distance, NAT and PAT, gateway redundancy, router on a stick
1:45–1:55	Break	24	10-minute break
1:55–2:35	VLANs, switching and spanning tree	25–35	VLANs, trunks, port security, loops and root, STP states, guards, MTU; activity and discussion
2:35–2:55	Wireless, physical and zones	36–44	Segmentation activity, zones, wireless bands, encryption, authentication, installation
2:55–3:00	Quiz & close	45–48	Quiz_Session_02.html, takeaways, checklist, preview

Materials and preparation

- Projector or screen with the Session 2 slide deck (02_Slides.pdf or 02_Slides.pptx).
- One computer or tablet per pair for Quiz_Session_02.html (any browser, no internet required).
- Printed or digital copy of the Student Notes for each candidate.
- Whiteboard or flipchart with three blank diagrams drawn ready: a switch and its table, a routing table with two sources for one destination, and a routed VLAN design.

- Optional: show a live MAC address table, a routing table and an adjacency table from a lab device if available.

Delivery guidance by slide

Slides 1–2 · Welcome and objectives (10 minutes)

Welcome, and set the frame. Session 1 gave the models and the numbers; today we put devices around them. The deck follows five chapters: infrastructure devices, cloud and routing foundations, route selection and translation, VLANs and spanning tree, and wireless with physical deployment. State the thread on slide 3 out loud and keep returning to it: every device is also a security boundary. Ask of each option, who does this protect, and from whom.

Slides 3–9 · Infrastructure devices (40 minutes)

Start with the two workhorses (slide 4). Frame against packet is the distinction everything else hangs on: a switch never reads an IP address. Walk the forwarding steps on the board, then the MAC flooding note, and make the security point that hub mode undoes segmentation at the port. Slide 5 adds the table and flooding in four steps: use the steps block as a live trace, one box at a time, and ask the room to predict the next action. Slides 6 and 7 cover the appliances and the traffic shapers. Land the sentence that matters: the firewall refuses by rule, detection reports after the fact, and prevention blocks in line. For proxies, draw two arrows, one hiding the client and one hiding the server. Slide 8 is a table; drive it as recall rather than reading, and check that candidates can pair each device with its protocol. Slide 9 is the discussion: run it for 6 minutes in pairs and 4 minutes of debrief, and use it to introduce defence in depth and the operational reality that an unread alert is not a control.

Discussion prompt (slide 9)

Which one stops an attack? The intrusion prevention system, because it is inline and can drop the packet. Why not rely on it alone? Signatures lag, tuning is imperfect and encrypted traffic limits inspection, so the firewall rule set, the detection capability and the response process all carry weight. Push the room on who reads the alerts.

Slides 10–16 · Cloud and routing foundations (25 minutes)

Chapter 2 turns to the cloud, then to routing. On slide 11, make the security consequence explicit: when a firewall becomes software, the management plane and the hypervisor become the target, and the patch cycle becomes an attacker's opportunity. Slide 12 walks the virtual private cloud: you define the address ranges, subnets and route tables; security groups are stateful and attached to a resource; network ACLs filter at the subnet boundary; a VPN gateway or a direct connection links back to the office. Slide 13 is the cloud model table; spend the time on the service models, because the exam asks where responsibility stops. Routing begins on slide 14 with static routes, administrative distance 1 and the stub network. Slide 15 introduces distance vector and the phrase routing by rumour: neighbours advertise their whole table, RIP measures hops, fifteen is the maximum and sixteen means unreachable. Slide 16 is OSPF, the open standard link state protocol: the same link state database on every router, the shortest path first calculation with self as root, cost following bandwidth, and area 0 as the backbone.

Slides 17–23 · Route selection and translation (30 minutes)

Slide 18 adds EIGRP and BGP. EIGRP is advanced distance vector with a feasible successor, so failover to the backup path needs no recalculation; BGP is the path vector protocol of the internet and selects on attributes and policy. Slide 19 is the administrative distance table: connected 0, static 1, EIGRP 90, OSPF 110, RIP 120. Drill the rule in order: administrative distance picks the source, the metric picks the path within that source, and the longest match rule wins overall. Slides 20 and 21 cover NAT and PAT. Keep the vocabulary straight: inside and outside give position, local and global give scope; PAT tells sessions apart by port number, which is why inbound services need a port forwarding rule. Slide 22 is first hop redundancy: a shared virtual IP address, HSRP active and standby, VRRP as the open standard, GLBP sharing the load, and preemption handing the role back. Slide 23 is router on a stick: subinterfaces, one 802.1Q tag per VLAN, and the bottleneck that comes with putting every VLAN on one link.

Slides 25–35 · VLANs, switching and spanning tree (40 minutes)

VLANs are containment, and this is the heart of the session. Slide 26 gives the fundamentals: broadcast domain, membership by port, hardware address or protocol, VLAN 1 as the default, and a management VLAN kept away from users. Slide 27 is the trunk and the tag: four bytes, a twelve-bit VID, just over four thousand usable VLANs, and the native VLAN crossing untagged. Explain both VLAN hopping methods here, switch spoofing and double tagging, and why tagging trunks consistently matters. Slide 28 is the design table; work it as a discussion rather than a table read, and land the rule: move VLAN 1 to an unused ID, prune unused VLANs and restrict the trunk allowed list. Slide 29 configures the port: speed and duplex agreeing, port security with sticky learning and violations, PortFast on an endpoint port, and BPDU Guard to shut a port that receives a bridge protocol data unit. Slides 30 to 33 are spanning tree. Work the loop problem first: a broadcast storm consumes the network in seconds and the MAC table destabilises. Then the root bridge election, the lowest bridge identifier, the root port choice and the blocked ports. Slide 31 is the port state table, with the convergence note: thirty to fifty seconds for classic STP, seconds for RSTP. Slide 32 adds the modern variants, 802.1w with alternate and backup port roles and discarding, 802.1s mapping groups of VLANs to instances, and per VLAN spanning tree. Slide 33 covers the attacks and the guards, then slide 34 is the worked election discussion. Slide 35 closes the chapter with the MTU: 1500 bytes, fragmentation or an ICMP message back to the sender, jumbo frames at about 9000 bytes, and the requirement that every device in the path agrees.

Discussion prompt (slide 34)

Equal priority, so the MAC address breaks the tie and the lower one roots. On the non-root switch the root port is the lowest-cost path: the gigabit link at cost four forwards, the hundred megabit link at cost nineteen blocks. If the gigabit link fails, the blocked port transitions and the network reconverges, in seconds on RSTP and thirty to fifty seconds on classic STP.

Slides 36–44 · Segmentation, wireless and physical (25 minutes)

Run the segmentation activity on slide 36 as a design exercise: 6 minutes in pairs and 4 minutes of debrief, with the DMZ, the blackhole and management isolation as the anchors, then extend into microsegmentation, isolating sensitive workloads rather than whole departments. Slide 37 lifts segmentation into zoning: the DMZ bracketed by an external and an internal firewall, blast radius and lateral movement, the three campus layers, and IoT devices on their own segment every time. Chapter 5 then leaves the cable behind. Cover the bands

on slide 39, the identifiers and modes on slide 40, and the encryption history on slide 41, where the lesson is what happens when a control is broken early. Slide 42 is enterprise authentication: 802.1X before joining, a central authentication server, Protected EAP inside a TLS tunnel, EAP TLS with certificates on both sides, and a guest network with its own name, its own VLAN and a bandwidth limit. Slide 43 covers access point management and antennas, with the site survey as the point: place them by measurement, not by eye. Slide 44 finishes in the room: a nineteen inch rack, 1.75 inches per rack unit, labelled cable and documented patch panel, locked racks, hot and cold aisles, a UPS, and the plain truth that physical access defeats every logical control.

Slides 45–48 · Quiz and close (25 minutes)

Run the quiz, review the low scores, then walk the seven takeaways and assign the checklist. Close on the instruction to read every device twice, once as plumbing and once as a boundary, and preview Session 3: the services that run on all of this, DNS, DHCP, NTP and SSH, and the ways each is attacked.

Common misconceptions and corrections

Misconception	Correction
"Switches route between networks"	Switches forward frames at layer 2; routers forward packets at layer 3 between networks
"A MAC address is software-assigned"	It is burned into the interface by the manufacturer, though it can be spoofed in software
"An IDS stops attacks"	An IDS only alerts; an IPS sits inline and blocks, and a firewall refuses by rule
"A forward proxy and a reverse proxy are the same thing"	A forward proxy hides and filters for clients; a reverse proxy fronts, caches and terminates TLS for servers
"NAS and SAN are interchangeable"	NAS is file level with NFS and SMB; a SAN is block level with Fibre Channel and looks like a local disk
"The cloud provider secures everything"	Service models shift responsibility, they do not remove it; on IaaS you still patch the operating system
"A static route is always safest"	It never adapts, so a failed link leaves traffic heading into a dead end
"16 hops means a very long path"	In RIP, 16 means unreachable: it is the value used to poison a route
"EIGRP and OSPF are the same kind of protocol"	EIGRP is advanced distance vector; OSPF is link state, so every router holds the whole map
"Administrative distance and metric do the same job"	Distance ranks the source of a route; the metric ranks paths from the chosen source
"NAT and PAT are two names for one thing"	PAT is a form of NAT that multiplexes many addresses onto one address by port number

Misconception	Correction
"A VLAN improves throughput"	VLANs are about segmentation and containment, not speed
"The native VLAN is safe by default"	VLAN 1 is the default native VLAN and a known attack surface on every trunk
"VLAN hopping is one technique"	Two methods: switch spoofing, which negotiates a trunk, and double tagging, which spoofs the native VLAN
"STP makes the network faster"	STP blocks loops for availability; RSTP and MSTP simply converge faster
"BPDU Guard stops root takeover"	Root Guard does that; BPDU Guard shuts a port that receives BPDUs at all
"A bigger MTU is always better"	Jumbo frames only help if every device in the path agrees; otherwise large packets vanish
"Hiding the SSID is a control"	It stops nothing; the name is visible to anyone watching the air
"WPA3 makes a weak password safe"	It resists offline cracking of a captured handshake, but the passphrase still has to be strong
"Physical security is somebody else's problem"	Physical access defeats every logical control, including full disk encryption at rest

Assessment check

- The objective list on slide 2 maps to the five chapters of this session and the course network-foundation strand.
- The HTML quiz tags every question with its topic area for traceable revision.
- The Wayground quiz (15 questions) gives an in-class, importable assessment of the same material.
- Segmentation and zoning appear in Domain 3 (architecture) and Domain 4 (operations) scenarios; virtualisation and shared responsibility appear in Domain 3 and Domain 4 cloud questions.
- Use the two discussions and the segmentation activity as formative checks: if the room cannot explain the longest match rule or the root bridge election, revisit before moving on.